EGYPT RAILWAYS MOBILE APPLICATION

OFFICIAL PROJECT DOCUMENTATION

TEAM 1

PROJECT REPOSITORY

https://github.com/EbramWagdy1/Egyptian-Railways.git

1-Proposal & Plan:

1.1 Project Overview & Objectives

- **Project Name:** Egypt Railways Mobile App
- **Technology Stack:** Flutter (Cross-platform for iOS & Android)

Overview: A modern, user-centric mobile application designed to digitize and streamline services for Egypt Railways passengers, including ticket booking, real-time schedule tracking, seat reservation, and customer support.

Core Objectives:

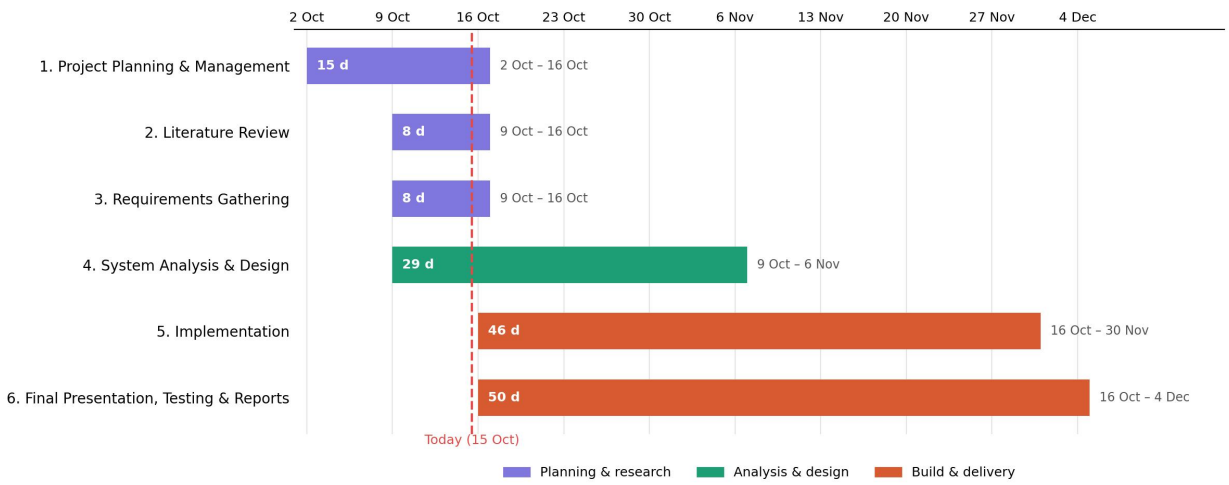
- Provide a seamless, fast, and secure booking experience.
- Reduce physical queues at station ticket windows.
- Deliver real-time train updates, delays, and schedule notifications.

Scope of Work

- **User Authentication:** Secure sign-up/login via phone number, email, and national ID verification.
- **Train Search & Schedule:** Filter by origin, destination, date, and train class (VIP, Air-conditioned, etc.).
- **Ticketing & Seat Selection:** Interactive seat map, booking workflow, and digital QR-code tickets.
- **Live Tracking:** Real-time status of trains and estimated time of arrival (ETA).
- **Bilingual Support:** Full Arabic and English interface.

1.2 Project Plan & Timeline

Project Gantt Chart (2 Oct – 4 Dec 2026)



Project Phase / Item	Start Date	End Date	Duration	Key Deliverables & Milestones
1. Project Planning & Management	2 Oct 2026	16 Oct 2026	15 Days	Project Proposal, Gantt Chart, Risk Assessment, KPI Setup
2. Literature Review	9 Oct 2026	16 Oct 2026	8 Days	Initial evaluation, literature research, feedback notes
3. Requirements Gathering	9 Oct 2026	16 Oct 2026	8 Days	Stakeholder analysis, User Stories, Functional/Non-Functional Specs
4. System Analysis & Design	9 Oct 2026	6 Nov 2026	29 Days	ERD schema, Architecture style, DFD, Sequence/Class diagrams, UI/UX Wireframes
5. Implementation (Source Code & Execution)	16 Oct 2026	30 Nov 2026	46 Days	Flutter codebase, REST APIs, Git repository, README, APK/IPA build
6. Final Presentation & Testing & Reports	16 Oct 2026	4 Dec 2026	50 Days	QA test cases, bug reports, User Manual, Final PPT, Video demo

1.3 Task Assignment & Roles

1. Mobile App Developer (Flutter)

- Develop Flutter UI components and application navigation.
- Implement mobile-side business logic and state management.
- Integrate RESTful APIs with the mobile application.
- Ensure responsive layouts and optimize application performance.

2. Backend Developer

- Design and implement RESTful APIs.
- Develop and manage the SQL database schema.
- Implement authentication and authorization services.
- Develop booking logic and reservation management.
- Integrate third-party payment gateways when available and authorized.

3. UI/UX Designer

- Create wireframes and high-fidelity Figma prototypes.
- Define color palettes, typography, and reusable design components.
- Design accessible Arabic and English interfaces with proper RTL/LTR support.
- Conduct usability testing and refine the user experience based on feedback.

4. QA & Testing Engineer

- Prepare and execute test cases and test scenarios.
- Perform unit, integration, and system testing.
- Document, track, and verify bug fixes.
- Conduct basic security and performance testing.
- Prepare test reports and support final quality assurance.

1.4 Risk Assessment & Mitigation Plan

Risk	Impact	Probability	Mitigation Strategy
High concurrent user traffic during peak booking hours	High	High	Implement scalable cloud architecture, API rate limiting, and Redis caching for schedule data.
Unavailability of official railway APIs or real-time data	High	Medium	Identify official data sources early, verify API access and permissions, and use clearly labeled mock data for prototype development when live data is unavailable
Double booking of train seats	High	Medium	Implement database transactions, enforce seat availability checks during booking confirmation, and use appropriate database constraints to prevent multiple confirmed bookings for the same seat and journey.
GPS live tracking inaccuracies	Medium	Low	Combine cell-tower positioning with station timestamp checkpoints for reliable estimations.

1.5 Key Performance Indicators (KPIs)

- **Response Time:** API response time under 500 ms for search and booking queries.
- **System Uptime:** 99.9% availability for the backend services and booking engine.
- **User Adoption Rate:** Target of 100,000+ active downloads within the first 3 months post-launch.
- **Crash-Free Users:** Maintain a stability rate of > 99% on both Google Play and App Store.

2- Literature Review

2.1 Feedback & Evaluation

The assessment methodology employs a multi-tier review framework designed to evaluate the platform across four core pillars: User Experience (UX), Technical Feasibility, Operational Reliability, and Architectural Scalability.

Assessment Pillar	Focus Area	Key Metrics & Evaluation Criteria
Core Functionality	End-to-End Ticketing Workflow	Verification of seamless search, selection, booking, payment execution, and ticket issuance workflows.
System Resilience	Edge-Case & Failure Handling	Graceful degradation under network dropouts, transaction rollbacks, and concurrent session limits.
Interface Architecture	Visual Design & Ergonomics	Compliance with platform design guidelines (Material 3 / iOS HIG), visual hierarchy, and contrast ratios.
Code Integrity	Software Architecture	Clean Code standards, modular design patterns (MVVM/Clean Architecture), robust error handling, and test coverage.

Strengths & Limitations Analysis

Strengths of Existing System

- **Core Utility:** Provides fundamental train search, class selection, and basic reservation capabilities.
- **Digital Generation:** Issues dynamic QR codes for digital ticket validation, reducing paper usage.
- **Historical Log:** Maintains a basic repository of past travel itineraries for registered users.

Critical Operational Bottlenecks

- **Payment Gateway Fragility:** Frequent unhandled timeouts during third-party payment processing result in debited accounts without ticket generation.
- **Authentication Bottlenecks:** Heavy reliance on standard SMS OTP causes severe onboarding and login friction due to carrier delivery latencies.
- **Static Telemetry:** Absolute absence of live train location updates, platform assignment changes, or unexpected schedule adjustments.
- **Connectivity Dependency:** Inability to load active tickets or display validation QR codes when operating without a live network connection.

2.2 Suggested Improvements

1. Booking Recovery & Transaction Status

- **Problem Statement:** Users experience payment debits without receiving a generated ticket due to connection drops during the gateway callback phase.
- **Proposed Solution:** Implement an idempotent transaction processing pipeline with automatic status polling, bi-directional webhook updates, and an immediate client-side 'Reconcile Transaction' button.
- **Feasibility & Priority:** High Feasibility | Priority 1 (Critical)

2. OTP & Authentication Resilience

- **Problem Statement:** Delays or drops in SMS delivery prevent users from logging in or finalizing time-sensitive ticket purchases.
- **Proposed Solution:** Integrate fallback authentication channels (WhatsApp Business API, IVR Voice Call verification, and optional passkey integration).
- **Feasibility & Priority:** High Feasibility | Priority 1 (Critical)

3. Smart Trip Reminders & Active Journey Mode

- **Problem Statement:** Critical information (platform, coach number, seat, departure time) is buried within dense ticket layouts.
- **Proposed Solution:** Introduce a dynamic 'Active Journey' dashboard card that automatically surfaces on travel day, displaying contextual travel information and real-time status.
- **Feasibility & Priority:** High Feasibility | Priority 1 (Critical)

4. Offline Ticket Access

- **Problem Statement:** Inability to display digital QR tickets inside subterranean terminals or areas with poor cellular reception.
- **Proposed Solution:** Implement secure local caching (encrypted via AES-256 with key management in hardware KeyStore/Secure Enclave) allowing full offline rendering of active ticket payloads and QR codes.
- **Feasibility & Priority:** High Feasibility | Priority 1 (Critical)

5. Real-Time Train Status & Delay Tracking

- **Problem Statement:** Passengers lack visibility into train location, operational delays, and actual arrival times.
- **Proposed Solution:** Deploy a live GPS telemetry mapping interface coupled with server-sent events (SSE) to push delay updates and estimated arrival times directly to the app.
- **Feasibility & Priority:** Medium Feasibility (Requires API Integration) | Priority 2 (High)

6. Destination Arrival Alert ('Don't Miss Your Station')

- **Problem Statement:** Passengers risk missing their destination stop, particularly on overnight or long-distance routes.
- **Proposed Solution:** Implement a background geofenced and time-calculated alert system that triggers high-priority visual, auditory, and haptic notifications prior to arrival.
- **Feasibility & Priority:** High Feasibility | Priority 2 (High)

7. Accessibility & In-App Support

- **Problem Statement:** The app lacks support for visually impaired users and offers inefficient customer support channels.
- **Proposed Solution:** Complete UI redesign adhering to WCAG 2.1 AA standards (high-contrast themes, screen-reader semantics) alongside an integrated self-service issue reporting system.
- **Feasibility & Priority:** High Feasibility | Priority 2 (High)

2.3 Final Grading Criteria

The project evaluation is structured across four key performance domains, totaling 100 Marks:

Category	Allocated Marks	Assessment Criteria & Detailed Rubric
1. Documentation & Analysis	25 Marks	• Depth and rigor of the Literature Review (7 Marks) • Quality of problem identification and domain analysis (6 Marks) • Comprehensive coverage of the Feature Improvement Matrix (6 Marks) • Architectural layout and systemic process diagrams (6 Marks)
2. System Implementation & Architecture	35 Marks	• Technical Architecture & Design Pattern application (10 Marks) • Production-grade UI/UX Implementation and WCAG compliance (10 Marks) • Robust State Management and Clean Code Practices (8 Marks) • Network resilience, caching, and offline fallback implementation (7 Marks)
3. Testing & Quality Assurance	20 Marks	• Automated Unit and Integration test coverage (7 Marks) • Empirical Usability Testing execution and results analysis (7 Marks) • Systematic edge-case handling (Network failure, error states) (6 Marks)
4. Final Presentation & Demonstration	20 Marks	• Live software demonstration and feature verification (8 Marks) • Technical presentation quality and clarity (5 Marks) • Team defense, domain knowledge, and response to technical Q&A (7 Marks)
TOTAL	**100 Marks**	**Comprehensive Academic Evaluation Criteria**

3. Requirements Gathering

3.1 Stakeholder Analysis

A. Passengers

Role & Impact: The end-users of the application. The overall success of the project relies heavily on their satisfaction and seamless user experience.

Key Needs & Expectations:

- User registration and authentication.
- The ability to create an account using their required personal information.
- Train search and booking capabilities.
- Generation and usage of electronic tickets.
- Receiving notifications and journey information.
- High usability and accessibility.

B. Train Conductors (Ticket Inspectors)

Role & Impact: Railway personnel responsible for checking and verifying passenger tickets on board the trains.

Key Needs & Expectations:

- A fast and reliable interface to verify electronic tickets.
- System availability and operational continuity to ensure the application functions consistently during journeys.

C. Station Agents

Role & Impact: Station staff who manage direct bookings and assist passengers on-site at the railway stations.

Key Needs & Expectations:

- A system module to process cash payment and ticket activation.
- Optimal system performance and response time to serve passengers quickly and prevent long queues at the station.

D. System Administrators

Role & Impact: The team responsible for managing the application, overseeing operations, resolving technical issues, and ensuring data security.

Key Needs & Expectations:

- Access to administrative functions to control and monitor the system.
- Tools for managing user accounts and providing customer support.
- Robust security, authentication, and access control mechanisms.
- Mechanisms for data integrity, regular backups, and system recovery.
- Ensuring user privacy and data protection.

3.2 User Stories

1. Passengers

- **User Registration and Authentication:** As a passenger, I want to create an account using my required personal information so that I can securely log in and access the system.
- **Train Search and Booking:** As a passenger, I want to search for trains and book my journey so that I can plan my travel efficiently.
- **Electronic Tickets:** As a passenger, I want to generate and use electronic tickets so that I do not have to carry physical paper tickets.
- **Notifications and Journey Information:** As a passenger, I want to receive notifications and journey information so that I stay updated on my trip status.
- **User Account and Support:** As a passenger, I want to manage my user account and access support so that I can keep my details updated and resolve any issues.
- **Cancellation Penalties:** As a passenger, I want to view cancellation penalties clearly so that I understand the rules and fees before canceling a booking.

2. Train Conductors (Ticket Inspectors)

- **Ticket Verification:** As a train conductor, I want a fast and reliable interface to verify electronic tickets so that I can efficiently check passengers on board.
- **System Reliability:** As a train conductor, I want the system to be consistently available during journeys so that I can perform my verification tasks without interruptions.

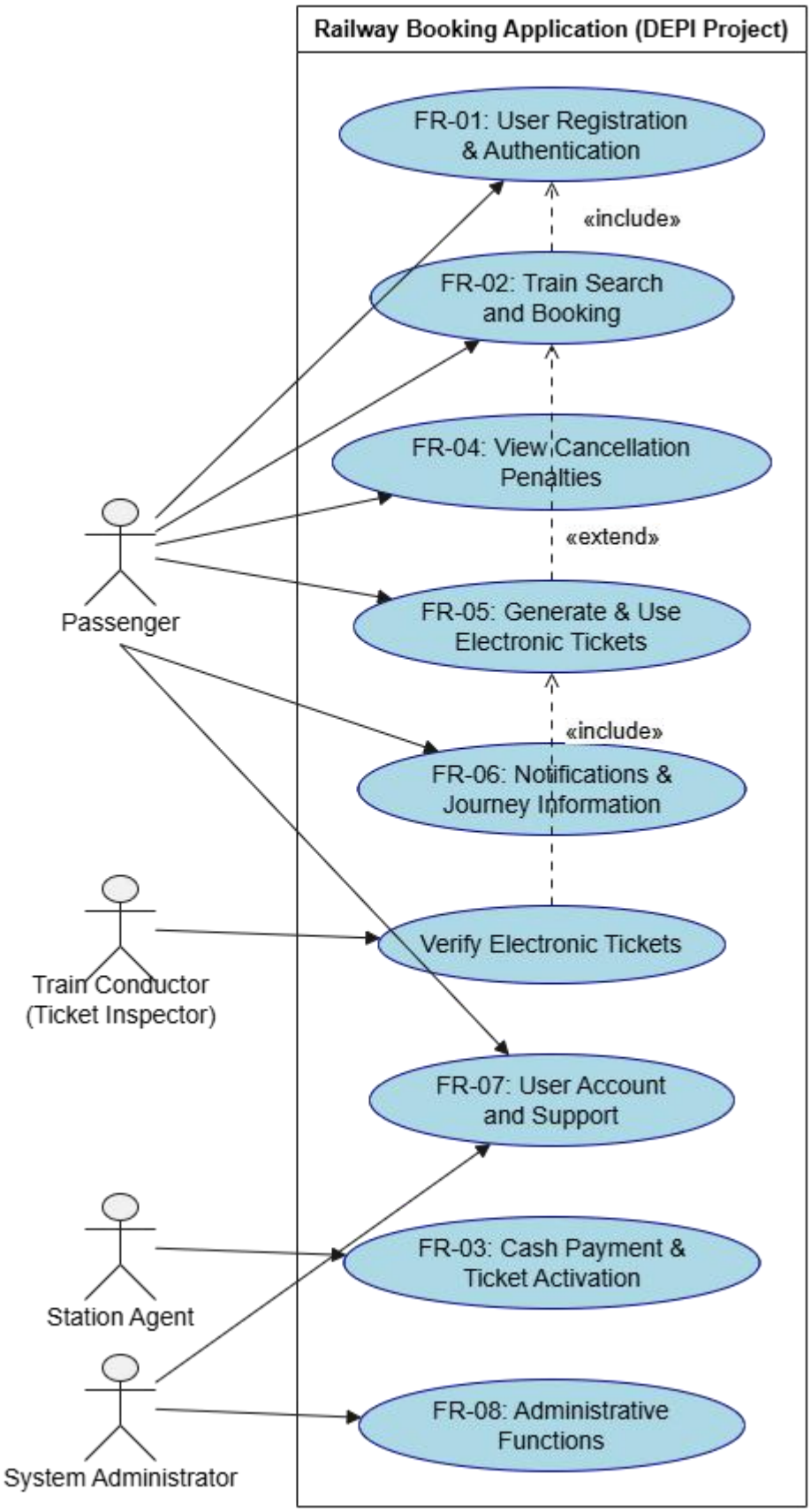
3. Station Agents

- **Cash Payment and Ticket Activation:** As a station agent, I want a module to process cash payments and activate tickets so that I can quickly serve passengers on-site and prevent long queues at the station.

4. Administrators

- **Administrative Functions:** As a system administrator, I want access to administrative functions so that I can oversee, manage, and control the overall railway system operations.

3.2 Use Case


Railway Booking Application (DEPI Project)
FR-01: User Registration & Authentication
«include»
FR-02: Train Search and Booking
FR-04: View Cancellation Penalties
«extend»
FR-05: Generate & Use Electronic Tickets
«include»
FR-06: Notifications & Journey Information
Verify Electronic Tickets
FR-07: User Account and Support
FR-03: Cash Payment & Ticket Activation
FR-08: Administrative Functions
Passenger
Train Conductor (Ticket Inspector)
Station Agent
System Administrator

3.3 Functional Requirements

- FR-01: User Registration and Authentication
- FR-02: Train Search and Booking
- FR-03: Cash Payment and Ticket Activation
- FR-04: Cancellation Penalties
- FR-05: Electronic Tickets
- FR-06: Notifications and Journey Information
- FR-07: User Account and Support
- FR-08: Administrative Functions

3.4 Non-Functional Requirements

- NFR-01: System Performance and Response Time
- NFR-02: Security, Authentication, and Access Control
- NFR-03: System Reliability and Booking Consistency
- NFR-04: System Availability and Operational Continuity
- NFR-05: Usability and Accessibility
- NFR-06: Platform Compatibility and Language Support
- NFR-07: Maintainability and Scalability
- NFR-08: Data Integrity, Backup, and Recovery
- NFR-09: User Privacy and Data Protection